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Synopsis on
AstraSafe: IOT Based
Women Safety Belt

Submitted in partial fulfilment of the requirements

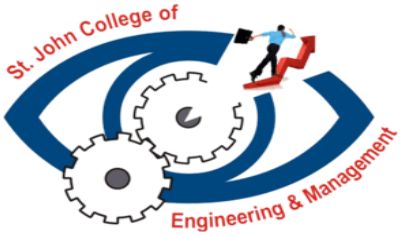
of the degree of

S.Tech
In
Computer Science and Engineering
(Data Science)
By

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CSE (Data Science) SJCEM



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI
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This is to certify that Roshni jha, Prachi Giri, Anushka Chauhan, Greeshma sarvankar are bonafide students of S. Tech in CSE (DS) Department, St. John College of Engineering and Management, Mumbai. They have satisfactorily completed the requirements of Hardware Report (Synopsis) as prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on AstraSafe IOT based women Safety Belt.

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Declaration

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ABSTRACT

Personal safety remains a paramount concern for women in modern urban and rural settings. Existing safety solutions often rely on smartphone applications that require manual interaction, which may not be feasible during a sudden physical threat or fall. This report presents the hardware design of AstraSafe, an IoT-Based Smart Women's Safety Belt. The wearable unit is built around an ESP32-C3 microcontroller board that is interfaced with a tactile SOS push button, an MPU6050 six-axis accelerometer/gyroscope for detecting falls or sudden motion, and a NEO-6M GPS module with an external antenna for capturing location coordinates. The modules are assembled on a breadboard using jumper wires: the MPU6050 communicates over the I2C bus, the GPS module over a UART serial port, and the SOS button is read through a digital input with a pull-up resistor. When the button is pressed or the motion pattern crosses a programmed threshold, the ESP32-C3 generates an emergency event along with the latest valid GPS coordinates and transmits it over its USB serial interface. The proposed hardware is compact, low-cost and expandable, and provides a foundation for future GSM/SMS, battery-powered and enclosure/PCB-based versions of the safety belt.

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Abbreviations and symbols

Abbreviation	Meaning
IoT	Internet of Things
SOS	Emergency distress signal
GPS	Global Positioning System
GSM	Global System for Mobile Communications
USB	Universal Serial Bus
I2C	Inter-Integrated Circuit
UART	Universal Asynchronous Receiver-Transmitter
GPIO	General Purpose Input/Output
NMEA	National Marine Electronics Association (GPS data sentence format)
SDA / SCL	Serial Data / Serial Clock lines of the I2C bus
MPU6050	Six-axis accelerometer and gyroscope sensor
MEMS	Micro-Electro-Mechanical Systems
NEO-6M	GPS receiver module family (u-blox)
SDG	Sustainable Development Goal
BOM	Bill of Materials

1. INTRODUCTION

1.1 Background

Personal safety technologies increasingly combine wearable sensors, location systems and microcontrollers to help users report emergencies. A wearable form factor such as a belt can provide quick access to an SOS control, while motion sensors can contribute to detecting possible falls or unusual movement. Embedding these functions in one compact electronic unit allows an alert to be generated without the user having to unlock or operate a phone.

This report focuses on the hardware of the AstraSafe safety belt. The system is built around the ESP32-C3 microcontroller, which reads a tactile SOS button, an MPU6050 accelerometer/gyroscope over the I2C bus, and a NEO-6M GPS module over a UART serial port. The components are assembled and wired on a breadboard using jumper wires, and the board is powered and programmed through its USB port. The GPS module can provide coordinates only when its antenna has a clear view of the sky and a valid satellite fix is obtained.

1.2 Problem Statement

Many safety solutions depend on the user opening a phone application or manually contacting someone during an emergency. Other prototypes use bulky Arduino-based assemblies with separate GSM and GPS boards that are costly and difficult to wear. There is a need for a compact, low-cost wearable hardware unit that combines a manual SOS trigger, motion-based fall detection and GPS location capture on a single microcontroller platform, with correct electrical interfacing (3.3 V logic, I2C and UART connections, and a debounced push button) between all the modules.

1.3 Need of the Project

Provide a fast, easily reachable manual SOS trigger on a wearable prototype.
Detect possible falls or sudden movement using a MEMS accelerometer/gyroscope.
Capture geographical coordinates using a GPS receiver with an external antenna.
Use a single low-cost, low-power microcontroller to integrate all sensors and inputs.
Create a modular hardware base that can later be extended with a GSM/SMS module, battery and enclosure.

1.4 Objectives

Select and study the hardware components: ESP32-C3 board, MPU6050, NEO-6M GPS module with antenna, tactile SOS button and resistor.
Interface the MPU6050 with the ESP32-C3 using the I2C bus and verify acceleration/gyroscope readings.
Interface the NEO-6M GPS module with the ESP32-C3 over UART and read NMEA location data.
Connect the SOS push button with a pull-up resistor and read it reliably through a GPIO pin.
Assemble the complete circuit on a breadboard with jumper wires and test the modules together.
Plan future hardware enhancements (GSM/SMS module, battery power, enclosure/PCB) without claiming that they are currently implemented.

1.5 Scope

The project covers the selection, interfacing, wiring and assembly of the electronic components of the AstraSafe safety belt on a breadboard prototype, together with basic functional testing of the SOS button, motion sensor and GPS module. The present prototype is powered through the USB port of the ESP32-C3 board and sends its output over USB serial. It does not currently include a GSM/SMS module, a battery pack, a buzzer, a custom PCB or a wearable enclosure. GPS accuracy depends on satellite reception, and the device is a supporting safety aid that cannot guarantee protection or emergency response.

1.6 Expected Outcomes

A working breadboard prototype combining ESP32-C3, MPU6050, NEO-6M GPS and an SOS push button.

Reliable manual SOS detection and motion/fall-event detection by the microcontroller.

GPS coordinates available to the microcontroller whenever a valid satellite fix is obtained.

A documented circuit diagram, pin mapping, test plan and roadmap for GSM/SMS, battery and enclosure enhancements.

2. LITERATURE REVIEW

2.1 Introduction

Research on wearable women's safety systems commonly combines an emergency trigger with GPS location, GSM/SMS communication and motion or other sensors. The main hardware design trade-offs include ease of activation, size and wearability, location reliability, false-positive handling, power consumption and cost. The following survey summarizes relevant published work identified for this report. Reported capabilities are those described by the respective publication summaries; they are not results of the present project.

2.2 Literature Review Table

Paper / Source	Year / Venue	Method / Focus	Finding / Relevance to proposed project	Additional details
The Role of IoT in Woman's Safety: A Systematic Literature Review	IEEE Access, 2023	Systematic review of IoT safety devices, sensors, communication and ML methods.	Identifies GPS/GSM and multiple sensors as common; highlights need for lower-interaction automatic alerts and improved accuracy.	Our prototype integrates the SOS button, MPU6050 and NEO-6M on one ESP32-C3 board.
Women Safety System using IoT	IEEE ICAAIC, 2022	Arduino, accelerometer, SIM808 GPS/GSM, push button and buzzer.	Describes button-triggered alerts and local alarm mechanisms.	We use a compact ESP32-C3 board instead of an Arduino and omit the GSM module at this stage.
A Smart friendly IoT device for women safety with GSM and GPS location tracking	IEEE ICECA, 2021	Wearable smart band using GPS/GSM and mobile connectivity.	Explores wearable location and emergency communication.	Our hardware captures location with a stand-alone NEO-6M and external antenna.
IoT based Unified approach for Women safety alert using GSM	IEEE ICICV, 2021	Wearable wristband with trigger, GSM, GPS, IoT and alert components.	Presents an integrated wearable alert concept.	We adopt the same trigger-plus-GPS concept using a tactile button and a motion sensor.
Design and Implementation of a GPS-GSM Based Women Safety Device for Combating	IEEE STI, 2023	GPS/GSM-based portable emergency device.	Describes sending location-based SMS to predefined contacts.	GSM hardware is outside the current prototype; GPS and motion sensing are interfaced.

Sexual Assaults				
Empowering Safety: Designing of an IoT based Women Protective System using AI enabled Smart Sensors	IEEE ACCAI, 2024	Sensor-enabled wearable safety system with GPS/GSM communication.	Explores smart sensors and location-based notifications.	We use a single MEMS IMU (MPU6050) with threshold-based motion sensing.
Women's Safety with Smart IoT	IJSAT, 2025	ESP32-based device using GPS, GSM, button and mobile app concepts.	Describes SMS and call notifications to configured contacts.	Shares the ESP32 platform and button/GPS design; cellular module is future work.
IoT-Based Women Safety Device Using Haversine Formula	IEEE ICREST, 2025	GPS/GSM and location calculations for nearest police station.	Explores location-aware emergency assistance.	Nearest-station calculation is not included; GPS coordinates are captured for context.
SOS-Enabled Wearable for Personal Security with GPS Tracking and Fall Detection	IEEE conference publication, 2025/2026 record	ESP32, SOS button, GPS/GSM and MPU6050 motion sensing.	Combines manual SOS and possible fall-detection events.	Closest to our hardware: ESP32, SOS button, GPS and MPU6050 motion sensing.
An Emergency Alert and Monitoring System using Compact IoT-based real-time women and child safety	IEEE ICSSS, 2025/2026 record	ESP32-CAM, GPS/GSM and connected alert features.	Describes compact alerts and evidence/location sharing.	Camera hardware is outside the scope of our prototype.
Women Safety Device with GPS Tracking & Alerts	IJCRT, 2025	Microcontroller, GPS/GSM, and emergency alert design.	Discusses portable location-enabled emergency alerts.	Our contribution is a low-cost three-input ESP32-C3 prototype with documented wiring.
Women Safety & Security System Using GPS and GSM Module	INDJECE, 2025	Arduino-based GPS/GSM emergency alert system.	Describes sending distress messages with location details.	Our design uses an ESP32-C3 with higher integration and built-in USB; GSM is future work.

2.3 Research Gap

Prior systems often emphasize direct GSM/SMS delivery, continuous tracking or mobile applications, and many are built from several separate boards (Arduino plus GSM/GPS shields) that are large and expensive for a wearable. A useful gap is to build a compact, low-cost and well-documented hardware unit around a small ESP32-C3 board that combines a manual SOS button, MPU6050 motion sensing and a NEO-6M GPS receiver, with clear wiring and pin mapping that other students can reproduce. The proposed hardware addresses this gap at prototype level; it does not replace a fully tested stand-alone emergency communication device.

2.4 Novelty / Innovation

Use of the compact ESP32-C3 RISC-V microcontroller with native USB, Wi-Fi and Bluetooth LE as the single controller of the safety belt.
Combination of three input sources (SOS button, MPU6050 motion data and GPS coordinates) on one low-cost breadboard prototype.
Clear documentation of the I2C, UART and GPIO connections, including the pull-up resistor for the SOS button.
A modular layout in which a GSM/SMS module, battery and enclosure can be added without redesigning the core circuit.
Low component count and low cost, which makes the design easy to reproduce and repair.

3. PROPOSED METHODOLOGY

3.1 Proposed System

The proposed system consists of a wearable sensing unit built around the ESP32-C3 microcontroller board. The tactile SOS button is connected to a GPIO input with a pull-up resistor, the MPU6050 is connected to the I2C pins (SDA and SCL), and the NEO-6M GPS module is connected to a UART port so that its NMEA sentences can be read by the microcontroller. All modules share a common 3.3 V supply and ground obtained from the ESP32-C3 board, which itself is powered through its USB port. When the SOS button is pressed or the motion data crosses a programmed threshold, the microcontroller creates an emergency event, attaches the latest valid GPS coordinates and sends it over the USB serial interface.

3.2 Working Principle

The device is powered through USB; the ESP32-C3 initializes its GPIO, I2C and UART peripherals.
The NEO-6M GPS module continuously sends NMEA location sentences to the ESP32-C3 over UART once it acquires satellites.
The MPU6050 provides acceleration and rotation values to the ESP32-C3 over the I2C bus.
The ESP32-C3 continuously monitors the SOS button input (active LOW through the pull-up resistor) with software debouncing.
When the SOS button is pressed, or the acceleration pattern crosses the programmed fall/sudden-motion threshold, an event message is created.
If a valid GPS fix is available, the latest latitude and longitude are attached to the event; otherwise the event is marked as 'no fix'.
The event message is transmitted over the USB serial interface of the ESP32-C3 for use by a connected computer or display.
Automatic SMS delivery through a GSM module is not part of the current hardware; it is a planned enhancement.

3.3 System Architecture/Block Diagram

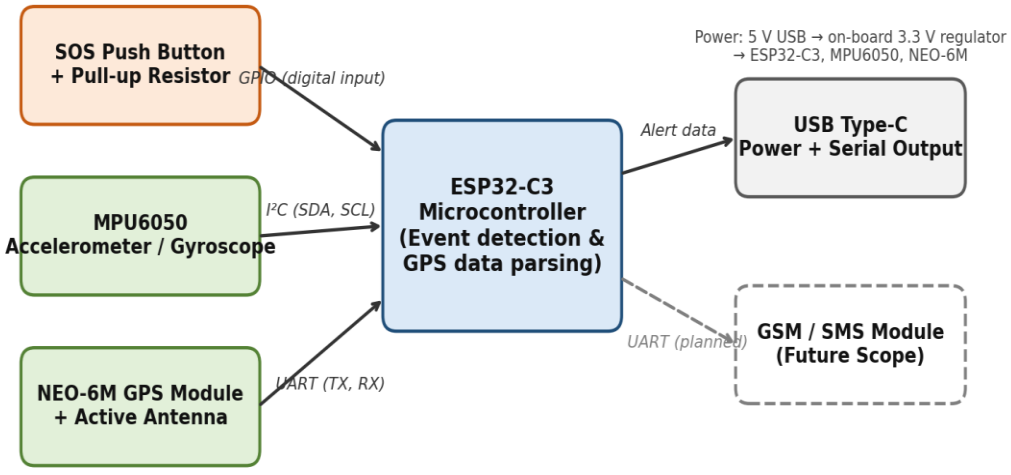


figure 3.1: Hardware block diagram of the AstraSafe safety belt

3.4 Flow Chart

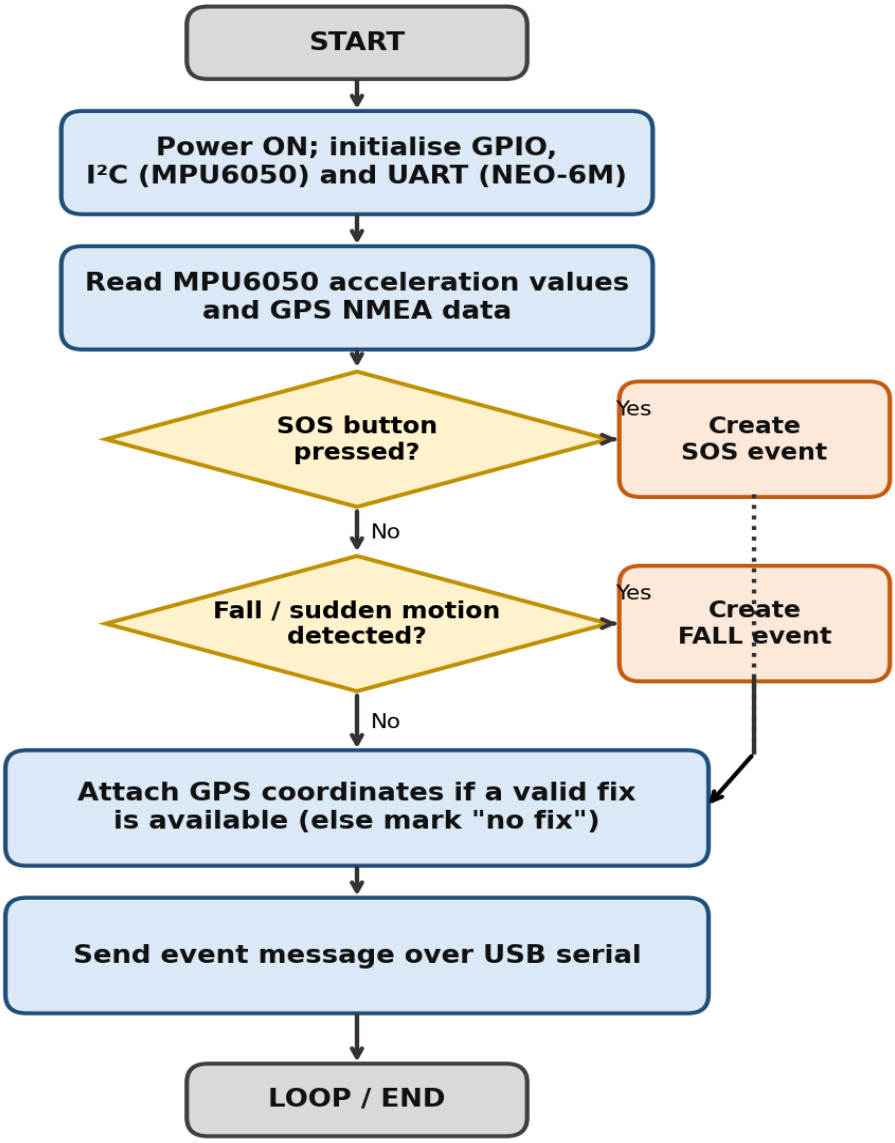


Figure 3.2: Flowchart of hardware operation

3.5 Methodology

1. Requirement analysis: define the inputs (SOS button, motion, GPS), the output (serial event message) and the operating limits of the prototype.
2. Component selection: choose the ESP32-C3 board, MPU6050, NEO-6M GPS module with antenna, tactile button and resistor after studying their datasheets and voltage levels.
3. Circuit design: prepare the pin mapping for I2C (MPU6050), UART (NEO-6M) and GPIO (SOS button) and draw the connection diagram.
4. Breadboard assembly: mount the modules and connect them with jumper wires, using a common ground and 3.3 V supply.
5. Module-wise testing: verify the I2C address of the MPU6050, the NMEA output of the GPS module and the response of the SOS button individually.
6. Integration: run all three modules together on the ESP32-C3 and confirm that events carry the correct sensor and location data.
7. Testing: test manual SOS, a simulated fall, GPS with and without a satellite fix, and loose-connection conditions.
8. Documentation: record photographs, wiring, test observations, limitations and the roadmap for GSM/SMS, battery and enclosure.

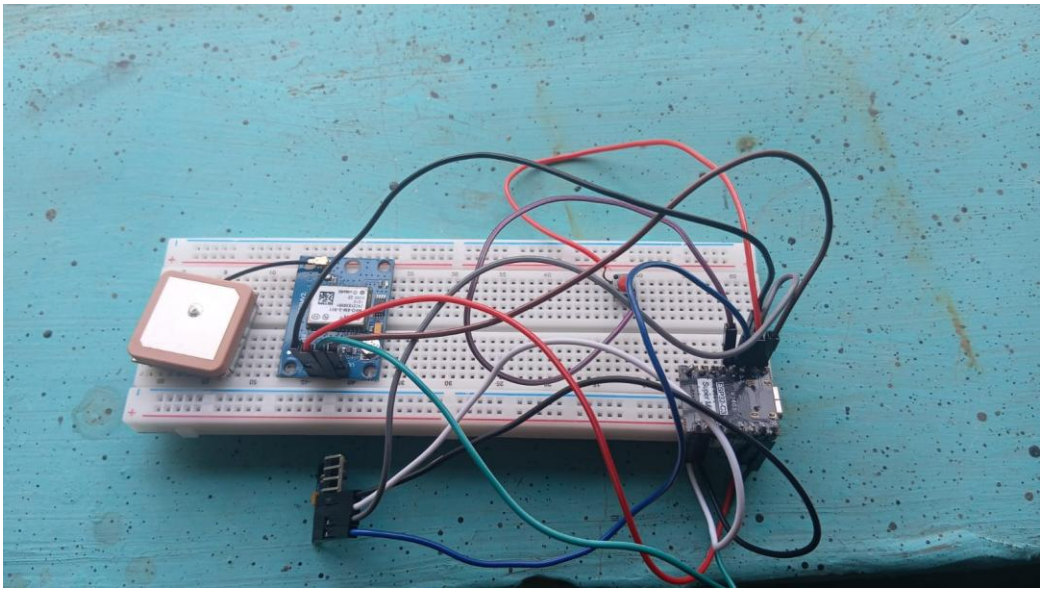


Figure 3.5.1: Prototype with ESP32-C3, NEO-6M GPS module and SOS button

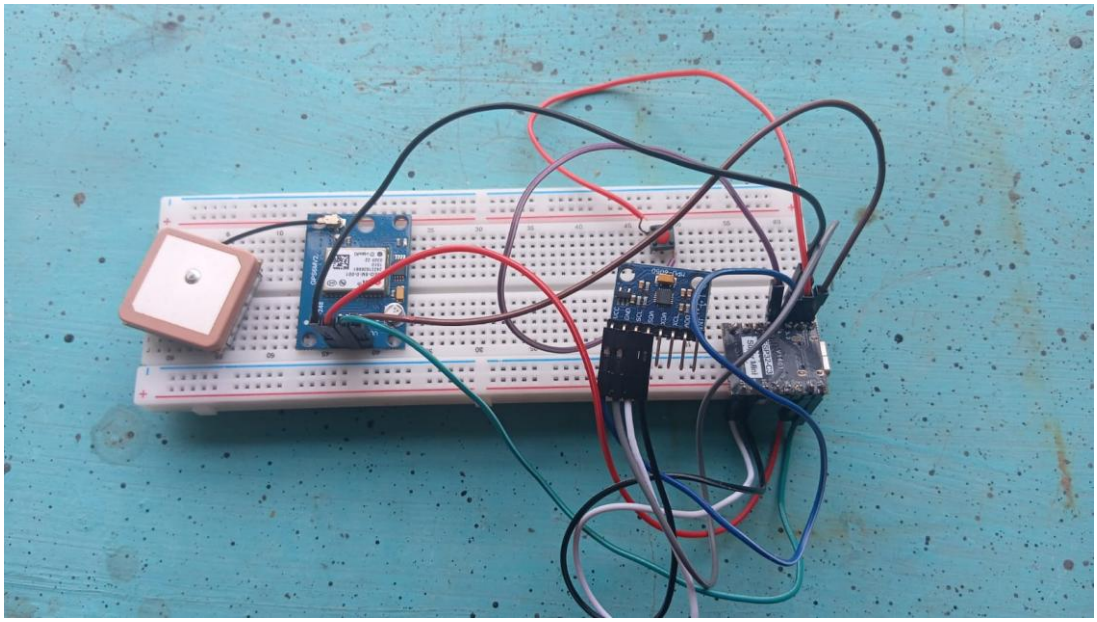


Figure 3.5.2: Complete prototype with MPU6050 sensor added

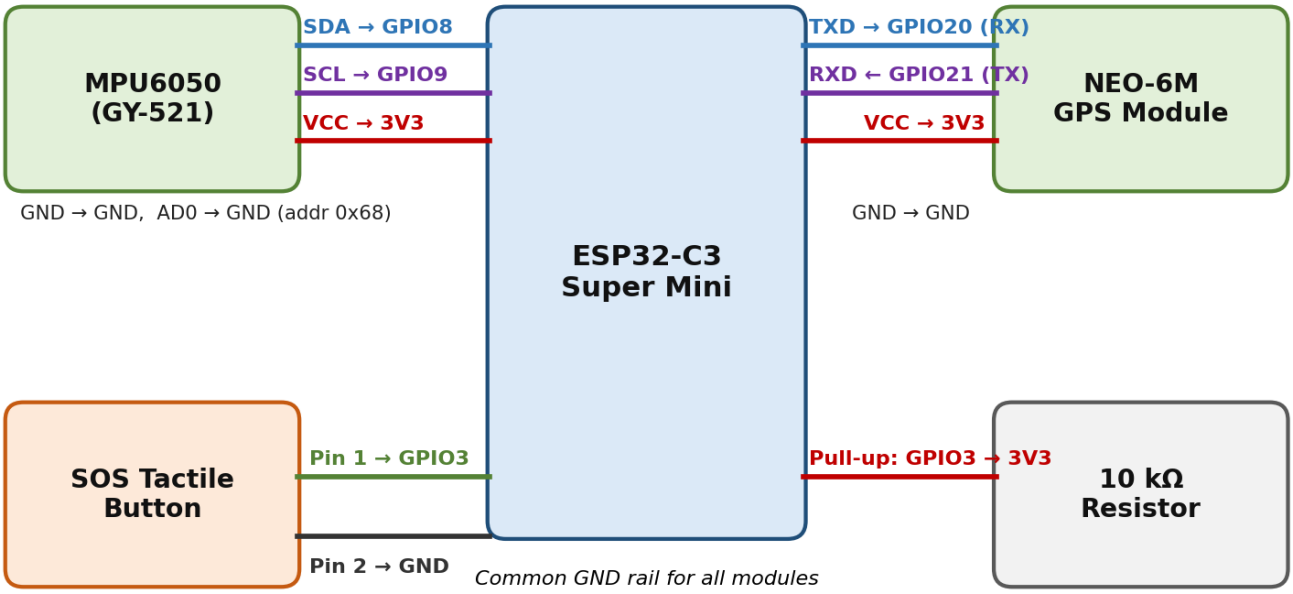


Figure 3.5.3: Circuit connection diagram of the hardware

3.6 Components / Materials

Component	Specification / Role	Quantity
ESP32-C3 microcontroller board (Super Mini)	32-bit RISC-V MCU up to 160 MHz, Wi-Fi + Bluetooth LE, native USB, 3.3 V logic; main controller [16]	1
MPU6050 accelerometer / gyroscope	6-axis MEMS sensor (3-axis accelerometer + 3-axis gyroscope), I2C interface, address 0x68; fall / motion detection [15]	1
NEO-6M GPS module with antenna	u-blox NEO-6M receiver with ceramic patch antenna, UART output (NMEA, 9600 baud default); location data [14]	1
Tactile SOS push button	Momentary push button for manual emergency trigger	1
Resistor kit (10 kΩ)	Pull-up resistor for the SOS button input	1 set
Breadboard	Solderless board for assembling and testing the circuit	1
Jumper wires	Male-to-male / male-to-female wires for connections	1 set
USB cable (Type-C)	Power supply and programming / serial output of the ESP32-C3	1

3.7 Pin Connections

Module / Pin	ESP32-C3 Pin	Purpose
MPU6050 VCC / GND	3V3 / GND	Power supply of the sensor
MPU6050 SDA	GPIO8 (SDA)	I2C data line
MPU6050 SCL	GPIO9 (SCL)	I2C clock line
NEO-6M VCC / GND	3V3 / GND	Power supply of the GPS module
NEO-6M TXD	GPIO20 (UART RX)	GPS data to microcontroller
NEO-6M RXD	GPIO21 (UART TX)	Configuration commands to GPS (optional)
SOS button (pin 1)	GPIO3	Digital input, active LOW
SOS button (pin 2)	GND	Ground return
10 kΩ resistor	Between GPIO3 and 3V3	Pull-up for the SOS button

4. PROJECT PLANNING

4.1 Work Breakdown Structure (WBS)

WBS ID	Task	Deliverable
1.0	Requirement analysis	Inputs, outputs and scope of the hardware
2.0	Component selection and procurement	Bill of materials and datasheet study
3.0	Circuit design	Pin mapping and connection diagram
4.0	Breadboard assembly	Wired ESP32-C3, MPU6050, NEO-6M and SOS button
5.0	Module-wise testing	Verified I2C, UART/NMEA and button input
6.0	Integration and testing	Combined prototype and test log
7.0	Documentation	Report, photographs and diagrams
8.0	Future enhancements	Plan for GSM/SMS, battery and enclosure

4.2 Gantt Chart / Semester Plan

Activity	Month 1	Month 2	Month 3	Month 4
Requirement study and literature review	✓	✓		
Component selection and procurement	✓	✓		
Breadboard assembly and wiring		✓	✓	
Sensor interfacing and module testing		✓	✓	
Integration and testing			✓	✓
Documentation and presentation				✓
GSM module, battery and enclosure (future scope)				Planned

4.3 Roles and Responsibilities

Team member	Suggested responsibility
Member 1	Hardware assembly and wiring on breadboard
Member 2	ESP32-C3 programming and sensor interfacing
Member 3	Testing, calibration and validation of modules
Member 4	Literature review, circuit diagrams and documentation

4.4 Risk Assessment

Risk	Impact	Mitigation
GPS has no valid fix (indoors / poor antenna view)	Location unavailable or inaccurate	Test outdoors with clear sky view; allow time for first fix; mark events as "no fix".
Loose jumper-wire / breadboard connections	Intermittent sensor or button failure	Re-seat wires, use short jumpers, test continuity; move to soldered/PCB version later.
Wrong voltage or wiring (3.3 V logic)	Damage to ESP32-C3 or modules	Check pin mapping and supply before power-on; use 3.3 V logic levels.
MPU6050 not detected on I2C	No motion data	Check SDA/SCL wiring, address (0x68) and pull-ups; run an I2C scanner.
False fall detection	Unnecessary alerts	Tune acceleration thresholds with controlled tests; add confirmation logic in future.
SOS button bounce or accidental press	Multiple or false events	Use pull-up resistor and software debouncing; consider a guarded button in the final design.
GSM / SMS module not included	No direct SMS to contacts	State the limitation clearly; integrate and test a GSM module in future work.

5. RESOURCES AND FEASIBILITY

5.1 Technical Feasibility

The prototype uses widely available and well-documented modules. The ESP32-C3 supports I2C, UART and GPIO on its 3.3 V pins, so the MPU6050 and NEO-6M GPS module can be connected directly without level shifters. The MPU6050 and NEO-6M are popular modules with extensive datasheets and library support. Technical feasibility depends on correct wiring, good jumper-wire contact, an antenna with a clear view of the sky for the GPS module, and a stable USB power supply.

5.2 Economic Feasibility

The prototype is designed to be low-cost by using common, off-the-shelf development modules. The ESP32-C3 board, MPU6050, NEO-6M GPS module, push button, resistors, breadboard and jumper wires together typically cost roughly ₹1,000 to ₹2,000 depending on the vendor. Optional future components such as a GSM module, battery pack and enclosure will add to the cost.

5.3 Operational Feasibility

The SOS button is intended to be simple and quick to press, and the whole unit is small enough to be mounted on a belt in a later version. In the present demonstration the circuit is on a breadboard and powered through USB, so it must remain connected to a power source or computer. A production wearable would require a battery, a soldered PCB or compact enclosure, and reliability testing.

6. EXPECTED RESULTS

6.1 Expected Results

- The ESP32-C3 should detect a press of the SOS button reliably and generate an SOS event without multiple false triggers.
- The MPU6050 should provide acceleration and gyroscope values over I2C, and a programmed fall or sudden-motion condition should generate a motion event.
- The NEO-6M GPS module should output NMEA data, and valid latitude/longitude should be available once the antenna obtains a satellite fix outdoors.
- All three modules should operate together on the ESP32-C3 without interfering with each other's communication.
- The event message with sensor and location data should be transmitted correctly over the USB serial interface.
- No real SMS delivery is expected until a GSM module is integrated and tested.

6.2 Performance Evaluation Plan

Metric	How to evaluate	Evidence to collect
SOS button response	Press the button repeatedly and count detected events	Serial output log and event count
MPU6050 data validity	Compare readings at rest and during controlled movements	Serial plotter / log of acceleration values
GPS time to first fix	Measure time from power-on to valid coordinates outdoors	Timestamped log and fix status
GPS availability	Test coordinate reception indoors, near windows and outdoors	Fix status and satellite count
False fall alerts	Compare triggered events with controlled test movements	Test log and threshold settings
Connection stability	Run the prototype continuously and move wires gently	Observation of resets or lost sensor data

6.3 Industrial Relevance

The hardware architecture demonstrates how a low-cost microcontroller, a MEMS motion sensor and a GPS receiver can be combined into a compact event-detection and location-capture unit. Similar designs can be adapted for personal safety, lone-worker assistance, elder-care fall alerts, asset tracking and industrial incident reporting after appropriate redesign, ruggedization and validation.

6.4 Expected Societal Impact

A discreet SOS control combined with motion sensing and GPS location may help users signal an emergency quickly. The system is intended as a supporting tool, not a guarantee of safety. Real-world use would require a battery-powered standalone design, reliable communication, accurate location and field testing.

6.5 Limitations

- The current prototype is built on a breadboard with jumper wires and is not yet a compact, rugged wearable.
- The unit is powered through USB; no battery or power-management circuit is included yet.
- GPS coordinates are available only when the receiver obtains a valid satellite fix, which may be slow or impossible indoors.
- Fall detection is threshold-based and may produce false positives or misses until tuned and validated.
- No GSM/SMS module is present, so the device cannot send messages to contacts by itself.
- There is no buzzer, enclosure or waterproofing in the present prototype.

7. SUSTAINABILITY & IMPACT

7.1 Sustainability

The design uses reusable development hardware and a small number of modules. The breadboard-based modular design allows each component to be replaced, reused or repaired individually. Future revisions should consider battery safety, low-power operation (for example, sleep modes of the ESP32-C3 and MPU6050), durable materials and responsible disposal of electronic components.

7.2 Environmental Impact

The prototype uses only a few electronic modules, but it still contributes to electronic waste if components are discarded improperly. Reuse of modules, repair, battery recycling and responsible e-waste disposal should be followed.

7.3 SDG Mapping

SDG	Possible relevance
SDG 5: Gender Equality	Supports exploration of technology-assisted personal safety for women.
SDG 9: Industry, Innovation and Infrastructure	Demonstrates integration of low-cost embedded hardware and sensors.
SDG 11: Sustainable Cities and Communities	May contribute to future connected safety-support systems.

7.4 Ethical and Privacy Considerations

- Use location data only for a clear emergency-related purpose and with appropriate permission.
- Do not describe a missing or unavailable GPS fix as a live location.
- Protect the privacy of any personal data stored or transmitted by the device.
- Avoid claims that the device guarantees protection or that an alert was delivered without evidence.

8.CONCLUSION

The hardware of the IoT-Based Smart Women's Safety Belt combines an ESP32-C3 microcontroller, a tactile SOS button, an MPU6050 accelerometer/gyroscope and a NEO-6M GPS module with antenna on a breadboard prototype. The design shows how I2C, UART and GPIO interfaces can be used together on a single compact board to detect a manual SOS, sense possible falls and capture location coordinates. The report also documents the limitations: breadboard-level construction, USB power, dependence on a valid GPS fix, threshold-based fall detection and the absence of a GSM/SMS module. Future work should focus on a battery-powered soldered/PCB version in a belt enclosure, tuned and validated fall detection, a buzzer for local alert, and tested GSM/SMS communication with trusted contacts.

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